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ORIGINAL ARTICLE

Cellular signatures in the primary visual cortex of phylogeny and placentation

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Abstract The long-held view that brain size can be used as an index of general functional capacity across mammals is in conflict with increasing evidence for phyletic differences in cellular organization. Furthermore, it is poorly understood how the internal cellular organization of the brain covaries with overall brain size variation. Using design-based stereology, we quantified glial cell and neuronal densities in the primary visual cortex of 71 mammalian species (spanning 11 orders) to test how those cellular densities are influenced by phylogeny, behavior, environment, and anatomy. We further tested cellular densities against mode of placentation to determine whether a relationship may exist. We provide evidence for cellular signatures of phylogenetic divergence from the mammalian trend in primates and carnivores, as well as considerably divergent scaling patterns between the primate suborders, *Strepsirrhini* and *Haplorrhini*, that likely originated at the anthropoid stem. Finally, we show that cellular densities in the mammalian cortex relate to the variability of maternal resources to the fetus in a species.

Keywords Evolutionary development · Primates · Comparative phylogenetics · Neocortex

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Introduction

Evidence for major phyletic differences in the cellular organization of mammalian brains has increased with advancing comparative histological studies. Variation among phylogenetic groups in the cellular density of cortical columns and in the morphology and biochemistry of neuronal phenotypes have been reported in rodents (Celio 1990; Beaulieu 1993; Skoglund et al. 1996), bats and Eulipotyphla (Valverde 1986), cetaceans (Garey et al. 1985; Haug 1987; Hof et al. 1992; Hof and Van der Gucht 2007; Butti et al. 2009), carnivores (Peters and Yilmaz 1993), and primates (Zilles et al. 1986; Peters and Sethares 1991; Preuss and Goldman-Rakic 1991a, c; Hof and Morrison 1995; Hof et al. 2000, 2001; Hof and Sherwood 2005). Further studies have reported differentiated expressions of calcium-binding proteins in different taxa (Glezer et al. 1992; Preuss and Kaas 1996; del Rio and DeFelipe 1997; Hof et al. 1999; Preuss 2001), qualitative and quantitative differences in cortical projection layers in primates (Preuss 2001), species-specific quantities of von Economo neurons (Nimchinsky et al. 1995, 1999; Butti et al. 2009; Hakeem et al. 2009; Allman et al. 2010), and diverse cellular scaling relationships in different mammalian orders (Herculano-Houzel et al. 2007; Azevedo et al. 2009; Sherwood et al. 2009; Herculano-Houzel 2010; Gabi et al. 2010). While homologous features exist across

mammalian brains (e.g., the basic arrangement of cortical laminae), universal modular architecture does not appear to exist and the argument to consider brain size as the most important index of general functional capacity common to all mammals, therefore, lacks strong support.

The number of neurons in the cerebral cortex is approximately determined by the number of progenitor cells, the duration of cell-division cycles, and the number of mitotic cycles during neurogenesis. Thus, the extended duration of progenitor cell division in macaque monkeys compared to mice results in an expansion of the surface area of the subventricular proliferative zone and a larger cortical mantle (Kornack and Rakic 1995, 1998; Kornack 2000; Rakic 2000). Once neuronal migration terminates, precursor cells divide to form astrocytes, a type of glial cell in the central nervous system (CNS). Astrocytes are the most abundant type of glia and are distributed isotropically in the cortical gray matter (Bushong et al. 2003; Ogata and Kosaka 2003; Nedergaard et al. 2003). Neurons co-cultured with astrocytes have been shown to achieve, through the production of trophic agents influencing synaptogenesis (Hatten et al. 1986; 1988; Muller 1993; Araque et al. 1999; Barres and Smith 2001; Hidalgo et al. 2001; Allen and Barres 2005), a sevenfold increase in synaptic efficiency compared to neurons raised in the absence of astrocytes (Pfrieger and Barres 1997; Ullian et al. 2001). After neurons in the developing brain have been surrounded by astrocytes and formed functional synapses, oligodendrocyte differentiation begins. These glial cells synthesize myelin, a lipid-rich membrane that ensheaths axons and increases the conduction velocity of electrical impulses. Many larger-brained phyla have independently evolved the ability to achieve efficient signal conduction speed with a smaller axon (Wang et al. 2008), and thus avoid the prohibitively space-consuming need for increasingly large axons in larger brains by wrapping the membrane capacitance with myelin synthesized by oligodendrocytes.

The implication that astrocytes may regulate the generation of new neurons (Song et al. 2002; Horner and Palmer 2003; Nedergaard et al. 2003), influence the development and synaptogenesis of those neurons (Pfrieger and Barres 1997; Kang et al. 1998; Haydon 2001; Ullian et al. 2001), monitor metabolic interactions at the synaptic cleft (Laming et al. 2000; Hertz et al. 2001), and generally required for dense synaptic networks to achieve advanced degrees of local modulation and control, as well as the need for oligodendrocytes to bypass axonal size constraints in increasingly large brains, suggests an evolutionary role for a relative increase in glial cells (hereon used to refer only to astrocytes and oligodendrocytes) in larger-brained species. The rate of this relative increase is unknown and its expression in specific brain regions has only begun to be tested (e.g., Sherwood et al. 2006a, b). Most studies of evolutionary variation in cortical organization have considered social or ecological factors (see Healy and Rowe 2007). However, it may also be important to consider what physiological processes external to the brain may affect the evolution of its organization. Because brain tissue is metabolically expensive, evolutionary variation must be constrained by the available energy (Aiello and Wheeler 1995; Hladik et al. 1999; Niven and Laughlin 2008). Mammals with relatively high metabolic rates do not produce relatively large-brained young, as would be expected if neonate brain size were determined by maternal metabolic rates (McNab 1989; Harvey and Krebs 1990; Capellini et al. 2010). Instead, species with relatively long gestational lengths produce relatively large-brained neonates (Cunnane and Crawford 2003; Barton and Capellini 2011; Capellini et al. 2011), which suggests that mode of placentation may influence neonatal growth (Martin 2008) and therefore be functionally related to evolutionary variation in cortical organization. The physiology of the placenta, which controls nutrient transfer to the developing fetus and shows remarkable evolutionary plasticity (Enders and Carter 2004), is likely to act as a major constraint on the evolutionary variation observed in neural tissue.

The current study compared neuronal and glial cell densities in the primary visual cortex in 37 primate species, 21 carnivore species, and 15 other mammalian species (spanning 11 orders). The aim of the study was to examine phylogenetic diversity in the quantitative histological organization of a homologous cortical area and test for relationships with evolutionary increases in metabolic activity, phylogenetic divergences, specializations in behavior, and developmental constraints. Furthermore, as recent evidence has shown that three placental types exist across the eutherian phylogeny (Lewitus and Soligo 2011), and that these types may be approximated by the degree of invasion of trophoblast cells into the uterus (Mossman 1987), we analyzed our data according to placental invasiveness to determine whether elements of placentation may influence variation in the mammalian cortex.

Materials and methods

Specimens

Samples of the left hemisphere of non-pathological post-mortem brains representing 73 mammalian species were used. All samples were from adult brains, except for *Trachypithecus francoisi* and *Pithecia pithecia*, which were from juveniles with brain sizes comparable to species-typical adult averages. Specimens from all collections were immersion-fixed with either 10% formalin or 4%

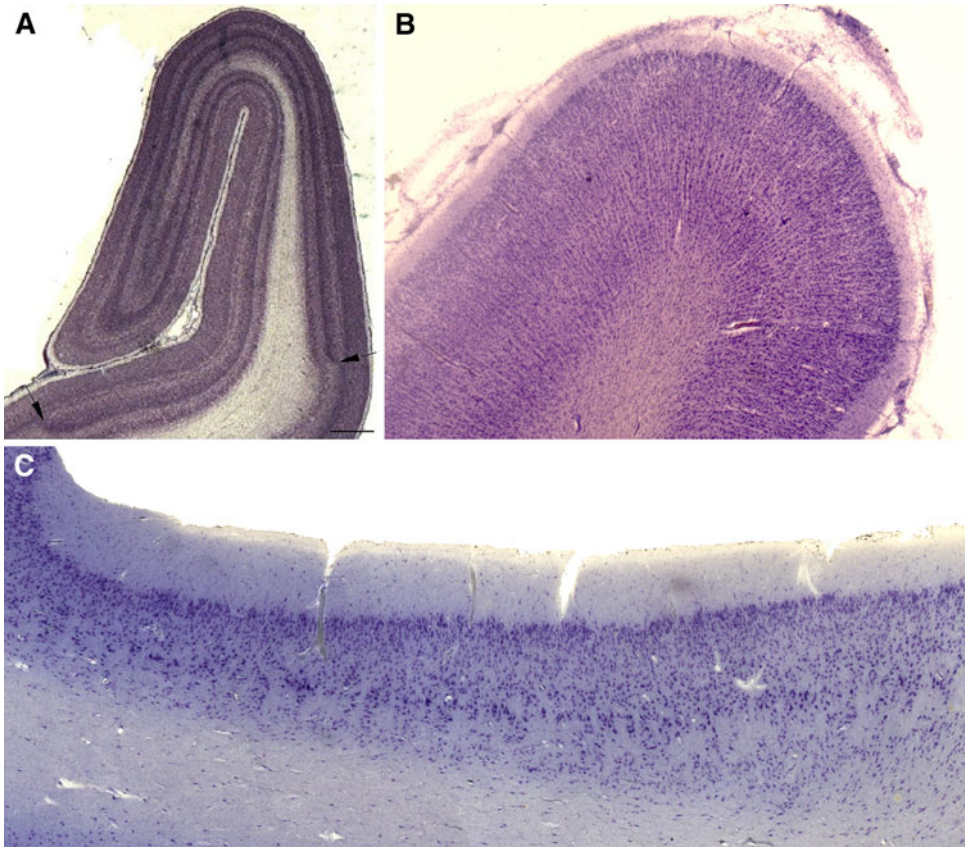


Fig. 1 The boundaries of V1 are clearly demarcated in primates (**a** *Callicebus moloch*) by the stria of Gennari (*arrows*), however, in carnivores (**b** *Ailurus fulgens*) the boundaries of V1 are identified by the relative thinning of layer IV in the flanking visual regions. In cetaceans (**c** *Megaptera novaeangliae*), V1 does not have a layer IV, like the rest of the neocortex, making it difficult to distinguish from adjacent cortical fields (see Hof et al. 2005 for details). *Scale bar* is 1 mm

paraformaldehyde, serially sectioned, and Nissl-stained with either cresyl violet or thionin in the context of this research or unrelated experiments. The original research reported herein was performed under guidelines established by the Animals Scientific Procedures Act (ASPA).

For the specimens originally prepared in the context of other investigations it was impossible to control for artifacts related to discrepancies in fixation length and postmortem delay. The recorded brain weights in our sample, nonetheless, do not show significant deviations from species-typical average fresh weights. When bivariate comparisons are made between densities of different cell types derived from the same individuals, comparable error from tissue shrinkage along both axes should be expected to affect the elevation of regressions, but not scaling exponents or residuals (see Sherwood et al. 2007).

Demarcation of V1

The mammalian V1 (Fig. 1a–c) is a readily identifiable architectonic area (Brodmann 1909; Inouye 1909; Holmes 1917; Talbot and Marshall 1941; Daniel and Whitteridge 1961; Schwartz 1977; Tootell et al. 1998), which we identified on the basis of its topological location and distinct appearance in materials stained for Nissl substance (Allman and McGuinness 1988; Hof and Morrison 1995; DeFelipe et al. 1999; Rosa 1999; Rosa and Tweedale 2005). A thin band of heavily stained tissue in cortical layer IVCα (the so-called stria of Gennari in humans) provided a landmark for identifying the area and its boundaries in primates (Boyd and Matsubara 2005). In non-primate species we used the Allen Reference Atlas (http://www.brain-map.org/), as well as sulci and cytoarchitectural landmarks identified in previous immunostaining studies of carnivores (Sherk 1986; Homman-Ludiye et al. 2010) and tree shrews (Conley et al. 1984; Wong and Kaas 2009) to demarcate V1. Studies of the marsupial cerebral cortex show that the visual regions flanking V1 are relatively thinner with a less prominent layer IV (Ashwell et al. 2008), characteristics observed in carnivores and primates and, therefore, likely to be conserved in mammals and useful as a general rule for identifying V1. The cetacean visual cortex, located between the paralimbic cleft (medially) and the superior lateral sulcus (laterally), however, is derived with respect to terrestrial mammals and lacks a layer IV (Garey et al. 1985; Hof et al. 2005; Hof and Van Der Gucht 2007; also see Furutani 2008 for a detailed report of cetacean V1 cytoarchitecture). The region of V1 sampled in all species was restricted to layers II–VI, since layer I was often poorly preserved in the older samples.

Cell counting

Cell counting was performed under brightfield microscopy using StereoInvestigator software (MBF Bioscience, Williston, VT, USA). The thickness of sections cut from the microtome ranged from 25–100 μm. Mounted section thickness was measured at the first and final counting site for each section using the microcator with a 63× objective and used to calculate volume estimates for cellular densities by dividing the total estimated cell population by the mounted section thickness. For each individual, a random starting section was selected in V1. Serial sections spaced at 300–400 μm were selected for analysis for each cell type. Boundaries of layers II-VI were outlined using a 10× objective, and a virtual 30 × 30-μm lattice of counting frames was randomly positioned on each slide to cover the sampled area with approximately 30 frames per section. Counting was performed under Koehler illumination using a 100× (NA 1.25, oil) objective—a 63× objective (NA 1.4, air) was used with one human individuals and one chimpanzee individual, and a 40× (NA 0.65, air) was used with the cetacean species, as the slides were too thick to allow for the working distance of higher power objectives. Since mounted section thickness varied, the disector thicknesses used also varied. A minimum 4-μm guard zone, defined as the space between the boundary of the tissue section and the part of the section used for counting, was set on either side of each section. Pilot tests were performed for each individual to determine the optimal size of the counting frame (approximately 2 particles per counting frame). Section thickness was measured at the first and final counting site for each section using the microcator. The coefficient of error (CE) was held below 0.08 ± 0.01 for all analyses (Gundersen and Jensen 1987; Gundersen et al. 1999; Slomianka and West 2005). Cellular density was calculated as the sum of neurons counted with the disectors, divided by the product of the sum of the disectors examined and the volume of the disector (Howard and Reed 1998).

Neurons were distinguished from non-neuronal cells by the presence of dark, coarsely stained Nissl substance in the cytoplasm, a large nucleus, a distinct nucleolus, ovoid shape, and lightly stained proximal segments of dendritic processes (Fig. 2). Excitatory and inhibitory neurons were not differentiated. Glial cells were expected to lack a conspicuous nucleolus and exhibit only a thin cytoplasmic rim around the nucleus. Astrocytes and oligodendrocytes were not differentiated and are classified simply as glia. Neither endothelial cells nor microglia were counted. The nucleolus of neurons and the body of glia were used for counting criteria.

Statistical analysis

Glia-neuron ratios were plotted against brain mass, body mass, encephalization quotient (EQ), and gestation length.

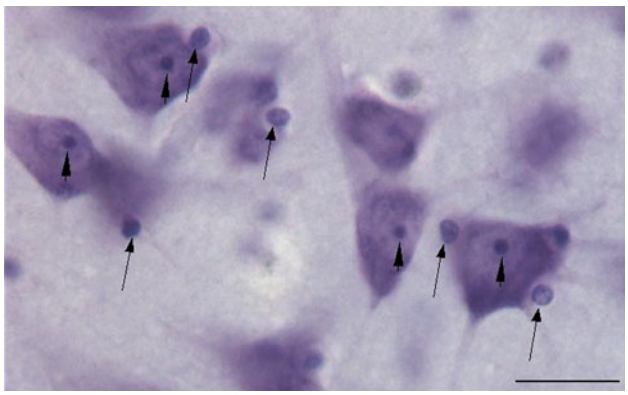


Fig. 2 Neurons (*arrowheads*) are identified by dark Nissl substance in the cytoplasm, lightly stained proximal segments of dendritic processes, and a prominent nucleolus, which was used in this study as the neuron's characteristic points for counting. Glia (*arrows*) are identified by the absence of a conspicuous nucleolus and less endoplasmic reticulum than neurons. *Scale bar* is 10 μm

In Primates only, glia-neuron ratio was plotted against neocortical volume, V1 volume, LGN volume, and axial and corneal diameter. When measurements of these additional variables were not available for the individuals studied, species mean data were taken from the literature (Ross and Kirk 2007; de Sousa et al. 2009, 2010). Data on reproductive traits were taken from the literature. Species were tested both as taxonomic groups—primates, carnivores, and eutherians (Table 1)—and according to mode of placentation—epitheliochorial, endotheliochorial, and haemochorial (Table 2)—defined here as the degree of invasion of trophoblast cells into the uterus (Mossman 1987). The eutherian taxonomic group consists of 32 species (spanning 13 phylogenetic orders), and includes: 14 species that are neither primates nor carnivores and a representative sampling of primate and carnivore species that was limited to three species in any taxonomic family. For this study, the group was considered representative of the superorder *Eutheria*. For non-parametric tests, the eutherians were stripped of primates and carnivore species, in order to avoid statistical non-independence. This stripped-down taxonomic group is referred to as shrew-to-whale (SW). Scaling exponents were determined by standard major axis (SMA) line-fitting based on log-transformed data. To determine whether observed glia-neuron ratios in humans deviated significantly from allometric expectations based on non-human taxa, least-squares (LS) prediction equations and prediction intervals were generated using species mean data and independent contrasts. The percentage differences between observed and predicted values were calculated as ((observed − predicted)/predicted) × 100). The same procedure was used to determine whether tarsiers deviated from expectations based on *Strepsirrhini* or *Haplorrhini*.

Table 1 List of species by taxonomic classification

Taxonomic group	Subgroup	Species	Taxonomic group	Subgroup	Species
Primates	*Strepsirrhini*	*Galago senegalensis*[a]	*Carnivora*	*Caniformia*	*Mustela nigripes*[a]
		Nycticebus coucang			*Neovison neovison*
		Lemur catta[a]			*Mephitis mephitis*
		Eulemur mongoz			*Taxidea taxus*
		Microcebus murinus			*Procyon cancrivorus*
		Cheirogaleus medius			*Procyon lotor*[a]
	Tarsiidae	*Tarsius bancanus*			*Nasua nasua*
		Tarsius syrichta[a]			*Bassaricyon gabbii*[a]
	Haplorrhini	*Callithrix geoffroyi*			*Potos flavus*
		Leontopithecus rosalia			*Ailurus fulgens*
		Saguinus oedipus[a]			*Zalophus californianus*
		Cebus capucinus			*Callorhinus ursinus*[a]
		Saimiri sciureus[a]			*Phoca vitulina*
		Aotus trivirgatus			*Ursus maritimus*[a]
		Callicebus moloch		*Feliformia*	*Canis lupus familiaris*
		Pithecia pithecia			*Canis latrans*[a]
		Alouatta caraya[a]			*Vulpes vulpes*
		Alouatta palliata			*Panthera pardus*[a]
		Ateles ater			*Felis catus*
		Macaca fascicularis (2)[a]			*Puma concolor*
		Macaca mulatta (2)			*Crocuta crocuta*[a]
		Macaca maura (5)			*Cynictis penicillata*
		Cercocebus torquatus	*Shrew-to-Whale (SW)*		*Dasyprocta leporina*[a]
		Mandrillus sphinx			*Lepus americanus*[a]
		Papio anubis (2)			*Cynocephalus volans*[a]
		Cercopithecus mitis			*Tupaia glis*[a]
		Cercopithecus nictitans			*Stenella coeruleoalba*[a]
		Erythrocebus patas (2)			*Tursiops truncatus*[a]
		Colobus angolensis[a]			*Megaptera novaeangliae*[a]
		Trachypithecus francoisi			*Sus scrofa*[a]
	Hominoidea	*Pongo pygmaeus* (2)[a]			*Scalopus aquaticus*[a]
		Pan paniscus			*Erinaceus europaeus*[a]
		Pan troglodytes (5)			*Sorex araneus*[a]
		Homo sapiens (6)[a]			*Trichechus manatus*[a]
		Gorilla gorilla (2)			
		Hylobates muelleri[a]			
		Symphalangus syndactylus			

The number of individuals sampled for each species is listed. Where no number is listed, only one individual was sampled

[a] Species included in Eutheria group

Independent contrasts were calculated using the PDAP:PDTREE module of Mesquite (Maddison and Maddison 2010) from a pruned mammalian phylogeny from Bininda-Emonds et al. (2007). Branch lengths were transformed according to Pagel (1992), as all the tips were contemporaneous. Prediction equations and intervals based on independent contrasts were generated by pruning humans from the tree, rerooting the tree at the last common ancestor of humans and other mammals, and then computing LS regressions and prediction intervals for independent contrasts (Garland and Ives 2000). Human predictions were generated from non-human data based on LS regressions, rather than SMA line-fitting, because the LS model was designed to generate predictions for y when x is known and produces residuals uncorrelated with the independent variable (Sokal and Rohlf 1995).

Data within taxonomic groups were tested for homogeneity of variance with Bartlett's test and for normality

Table 2 List of species by placental characterization

Haemochorial	Endotheliochorial	Epitheliochorial
Callithrix geoffroyi	*Tupaia glis*	*Lemur catta*
Leontopithecus rosalia	*Felis catus*	*Eulemur mongoz*
Saguinus oedipus	*Panthera pardus*	*Microcebus murinus*
Cebus capucinus	*Puma concolor*	*Cheirogaleus medius*
Saimiri sciureus	*Cynictis penicillata*	*Galago senegalensis*
Aotus trivirgatus	*Crocuta crocuta*	*Nycticebus coucang*
Callicebus moloch	*Canis lupus familiaris*	*Manis gigantea*
Pithecia pithecia	*Canis latrans*	*Sus scrofa*
Alouatta caraya	*Vulpes vulpes*	*Stenella coeruleoalba*
Alouatta palliata	*Ailurus fulgens*	*Tursiops truncatus*
Ateles ater	*Ursus maritimus*	*Megaptera novaeangliae*
Macaca fascicularis	*Mephitis mephitis*	
Macaca mulatta	*Bassaricyon gabbii*	
Macaca maura	*Mustela nigripes*	
Cercocebus torquatus	*Taxidea taxus*	
Mandrillus sphinx	*Neovison neovison*	
Papio anubis	*Procyon cancrivorus*	
Cercopithecus mitis	*Procyon lotor*	
Cercopithecus nictitans	*Potos flavus*	
Erythrocebus patas	*Nasua nasua*	
Colobus angolensis	*Phoca vitulina*	
Trachypithecus francoisi	*Callorhinus ursinus*	
Pongo pygmaeus	*Zalophus californianus*	
Pan paniscus	*Trichechus manatus*	
Pan troglodytes	*Cynocephalus volans*	
Homo sapiens		
Gorilla gorilla		
Hylobates muelleri		
Symphalangus syndactylus		
Tarsius bancanus		
Tarsius syrichta		
Dasyprocta leporina		
Erinaceus europaeus		
Scalopus aquaticus		
Sorex araneus		
Lepus americanus		
Crocuta crocuta		

with the Shapiro–Wilk's W test. Differences in distributions between taxonomic groups were tested using a two-sample Kolmogorov–Smirnov goodness-of-fit test. Differences in glia-neuron ratios and cellular densities between different taxonomic groups were calculated using Kruskal–Wallis sum rank and comparison tests with a repeated-measures design. When the population of the variable was normally distributed, ANOVAs were used to determine within- and between-group differences in glia-neuron ratio. Statistical significance for all analyses was set at $\alpha = 0.05$ (two-tailed). All analysis was performed in *R* with our own code and the packages *SMATR* (Warton et al. 2006) and *relaimpo* (Groemping 2006, 2007).

Results

We first compared cellular densities among carnivores, primates, and eutheria. The primate data were then divided into taxonomic subgroups to test whether cellular scaling relationships were conserved in primates. These subgroups were also used to determine if (1) humans and other apes

deviated from predictions made based on Old and New World monkeys and (2) tarsiers were better affiliated with *Strepsirrhini* or *Anthropoidea* in the context of glia and neuron scaling in V1. Finally, we compared cellular densities among placental groups to test whether mode of placentation, compared to behavior or phylogeny, had an influence on cellular densities in the primary visual cortex (V1).

Cellular scaling patterns in mammals

SMA exponents for species mean data and independent contrasts are presented in Table 3. The Eutheria group consists of 31 species, spanning 13 orders, and is considered a representative set of Eutheria. The eutheria group includes the 14 species sampled that are neither primates nor carnivores, as well as a representative sampling of primate and carnivore species. For non-parametric tests the eutheria group was stripped of its primate and carnivore species, in order to avoid statistical non-independence when comparing the carnivores and primates with other species (Fig. 3). The stripped-down group is referred to as Shrew-to-Whale (SW).

Glial cell density was found to scale against neuronal density with a significant positive exponent for carnivores and eutheria, as revealed by analysis of both species mean data and independent contrasts. For carnivores, the 95% CI of the species mean data included isometry. In primates, glial cell density scaled against neuronal density with a significant positive exponent only for independent contrasts, suggesting that the species mean data were obscuring grade shifts among major taxonomic groups within the primate data set (Garland et al. 1993). Applying weighted least-squares for heteroscedasticity in neuronal density ($K^2 = 7.054$, $p = 0.029$), line-fitting through the origin of LS regression slopes for glial cell density on neuronal density for independent contrasts revealed statistically different slopes for carnivores (LS slope = 0.789, lower CI = 0.658, upper CI = 0.920, $R^2 = 0.365$, $p = 0.037$) and primates (LS slope = 0.536, lower CI = 0.420, upper CI = 0.652, $R^2 = 0.288$, $p = 0.049$), and a significant allometric slope for eutheria (LS slope = 0.541, lower CI = 0.390, upper CI = 0.692, $R^2 = 0.720$, $p = 0.000$). The SMA plots of glial cell density against neuronal density displayed significantly different *y*-intercepts for carnivores (*y*-intercept = −0.343, lower CI = −0.678, upper CI = −0.008; $p = 0.049$) and eutheria (*y*-intercept = 1.05, lower CI = 0.754, upper CI = 1.346; $p = 0.000$).

Cellular scaling patterns within primates

Since the SMA exponent for glial cell density on neuronal density for *Haplorrhini* was only significant for independent contrasts, the extreme studentized deviate method and Cook's distance (set at 50%) were used to identify orangutans ($Z = 3.34$) and gorillas ($Z = 3.20$) as significant outliers. Thus, the ape lineage (*Hominoidea*) was removed from the *Haplorrhini* data set, as its constituents imposed too great an influence on the regression. Statistically different scaling exponents for glial cell density on neuronal density were then generated for *Strepsirrhini* (SMA = 0.715, lower CI = 0.477, upper CI = 0.998, $R^2 = 0.335$, $p = 0.011$) and Old and New World monkeys (SMA = 1.544, lower CI = 1.22, upper CI = 1.97, $R^2 = 0.219$, $p = 0.028$) for species mean data, which also held for independent contrasts.

The species mean data and independent contrasts for Old and New World monkeys were used to generate log–log LS regression predictions of glial cell density on neuronal density, and glia-neuron ratio on brain mass. The values for apes for glia-neuron ratio as a function of brain mass, but not for glial cell density as a function of neuronal density (Table 4), fell within the 95% prediction intervals (PI) of the LS regression for independent contrasts (Fig. 4a, b).

When the species mean data and independent contrasts from the non-human apes were used to generate log–log LS regression predictions of glial cell density on neuronal density, and glia-neuron ratio on brain mass, body mass, and gestation length, the observed human values for glial cell density against neuronal density were significantly greater than expected (paired-samples *t* test: $t = 9.02$, $p = 0.009$) and the observed human values for glia-neuron ratio on brain mass were significantly less than expected (paired-samples *t* test: $t = 10.07$, $p = 0.012$) for independent contrasts (Table 5). The observed values for glia-neuron ratio on body mass and gestation length fell within the 95% PI of the LS regression for non-human ape species mean data and independent contrasts.

Similarly, log–log regression predictions were generated for *Tarsius bancanus* and *Tarsius syrichta* from data based on Old and New World monkeys and *Strepsirrhini*. For both tarsier species, the observed glial cell density values fell within the 95% PI of the predicted values based on neuronal density scaling in Old and New World monkeys; but the observed glia-neuron ratio values fell within the 95% PI based on brain mass in *Strepsirrhini* (Fig. 4a, b).

Cellular scaling patterns among placental groups

Significantly different SMA exponents for glial cell density against neuronal density were found between the epitheliochorial (slope = −0.229, lower CI = −0.646, upper CI = 0.188, $R^2 = 0.859$, $p = 0.000$) and both the endotheliochorial (slope = 1.216, lower CI = 0.973, upper CI = 1.459, $R^2 = 0.346$, $p = 0.043$), and haemochorial (slope = 1.240,

Table 3 Slope estimates and correlation coefficients for scaling relationships between cell densities in V1 and anatomical variables

Taxa	Dependent variable	Independent variable	Species mean data					Independent contrasts				
			RMA	R^2	Lower 95% CI	Upper 95% CI	p	RMA	R^2	Lower 95% CI	Upper 95% CI	p
Eutheria ($n = 39$)	Glia-cell density	Neuronal density	0.733	0.099	0.578	0.928	0.035	1.035	0.140	0.880	1.231	0.000
	Gila-neuron ratio	Brain mass (g)	0.386	0.575	0.305	0.489	0.014	0.375	0.603	0.297	0.529	0.001
		Body mass (kg)	0.311	0.726	0.249	0.387	0.000	0.319	0.691	0.239	0.440	0.003
		Gestation length (d)	0.347	0.743	0.288	0.417	0.791	0.485	0.398			0.008
		EQ	−0.232	0.871	−0.9191	−0.282	0.010	−0.452	0.444	−0.362	−0.588	0.633
Camivores ($n = 21$)	Gila cell density	Neuronal density	1.051	0.003	0.691	1.599	0.049	1.221	0.102	0.928	1.392	0.028
	Glia-neuron ratio	Brain mass (g)	0.419	0.583	0.284	0.619	0.009	0.568	0.321	0.449	0.802	0.013
		Body mass (kg)	0.308	0.741	0.205	0.462	0.022	0.382	0.604	0.287	0.527	0.026
		Gestation length (d)	0.999	0.000	0.679	1.472	0.008	1.702	0.281	1.395	2.178	0.087
		EQ	1.425	0.117	0.900	2.255	0.113	−1.385	0.099	−1.108	−1.800	0.531
Primates ($n = 37$)	Glia cell density	Neuronal density	1.073	0.005	0.759	1.518	0.101	1.256	0.082	1.118	1.494	0.000
	Glia-neuron ratio	Brain mass (g)	0.333	0.660	0.236	0.469	0.113	0.525	0.333	0.415	0.741	0.208
		Body mass (kg)	0.274	0.766	0.196	0.384	0.025	0.377	0.568	0.298	0.531	0.318
		Gestation length (d)	1.635	0.210	1.145	2.334	0.42	2.006	0.367	1.505	2.769	0.435
		EQ	−0.981	0.000	−0.688	−1.399	0.150	0.858	0.024	0.686	1.116	0.606
		Axial diameter (mm)	2.167	0.421	1.474	3.184	0.814	−2.138	0.412	−1.604	−2.951	0.763
		Corneal diameter (mm)	2.380	0.493	1.623	3.489	0.620	−2.263	0.470	0.398	0.621	0.109
		V1 volume (cm^3)	0.417	0.502	0.274	0.637	0.330	0.457	0.443	0.388	0.544	0.526
		Neocortex volume (cm^3)	0.426	0.554	0.291	0.624	0.010	0.499	0.380	0.424	0.594	0.392
		LGN volume (cm^3)	0.506	0.359	0.329	0.780	0.320	0.464	0.440	0.380	0.594	0.406

lower CI = 1.004, upper CI = 1.476, $R^2 = 0.456$, $p = 0.003$) independent contrasts (Fig. 5; Table 6). ANOVA revealed a significant main effect of placental group on glia-neuron ratio ($F_{2,80} = 14.25, p = 0.000$). Bonferroni *post hoc* comparisons of mean glia-neuron ratio showed that the epitheliochorial (1.10 ± 0.35) and haemochorial (0.51 ± 0.21) groups differed significantly.

Discussion

By examining allometric scaling relationships across taxa, it is possible to gain insight into the functional implications of changes in brain size (Hawkins and Oszewski 1957; Friede and Van Houten 1962; Tower and Young 1973; Herculano-Houzel et al. 2007). Previous studies of species mean data sampled from a range of mammals have demonstrated that neuronal density in the neocortex scales with brain mass with an exponent of approximately −0.3 (Tower 1954; Prothero 1997; Sherwood et al. 2007). Our results produced similar scaling exponents for neuronal density on brain mass for eutheria (−0.376 ± 0.61), suggesting that cortical neuronal density is limited by architectural constraints imposed by brain mass. In line with these results is a model predicting that a constant average percent of interconnectedness among neurons cannot be physically sustained with increasing gray matter volume, and therefore the reach of processing networks does not keep pace with brain size variation (Changizi 2001). However, the steeper slopes observed in carnivores (−0.437 ± 0.50) and, especially, primates (−0.531 ± 0.23) in our study suggest that neuronal density may become relatively disassociated from brain mass in certain taxa or when comparisons are made among close phylogenetic relatives (see Sherwood and Hof 2007). Factors underlying neuronal density, such as glial cell density, may reflect

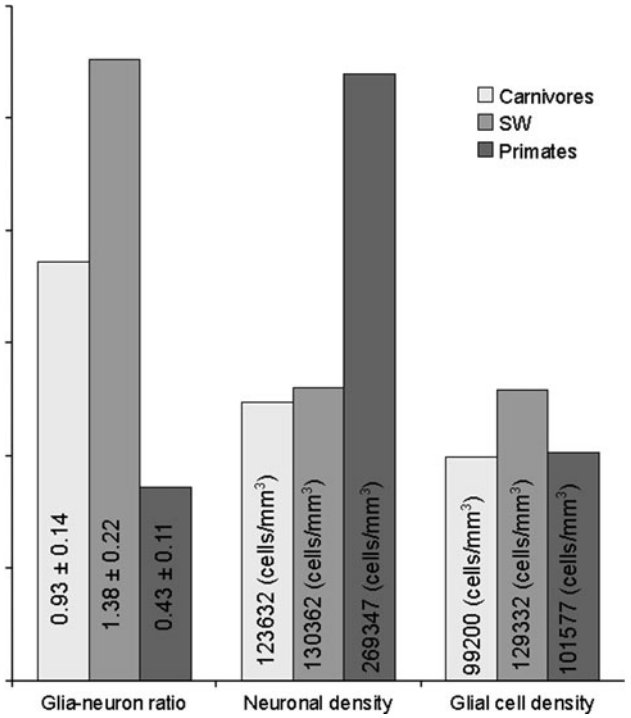


Fig. 3 Bar representations of Kruskal–Wallis sum rank and multiple comparison tests showing significant differences among carnivores, SW, and primates for glia-neuron ratio ($\chi^2 = 28.71$, $p = 0.000$) and between SW and both carnivores and primates for neuronal density ($\chi^2 = 27.36$, $p = 0.000$). No significant differences are shown for glial cell density ($\chi^2 = 1.24$, $p = 0.541$). Two-sample Kolmogorov–Smirnov tests for normality of distribution (bootstrap = 10,000) showed significant differences between carnivores and primates for glia-neuron ratio ($D = 0.688$, $p = 0.000$) and neuronal density ($D = 0.675$, $p = 0.000$), and between SW and primates for glia-neuron ratio ($D = 0.765$, $p = 0.000$) and neuronal density ($D = 0.659$, $p = 0.001$). Raw values for glia-neuron ratio and neuronal and glial cell density are displayed within the *bars*. *Bar* representations are not shown to scale

brain region-specific evolutionary adaptations within phylogenetic lineages that result in such deviations. Our results only partially support the claim that larger-brained species have increased glia-neuron ratios to pay for the increased costs of neuronal activity that come with modulating expansive arbors (Laming et al. 2000; Hertz et al. 2001; Harrison et al. 2002). Across all species in our sample, the mammalian area V1 displays a significant scaling relationship between glial cell density and neuronal density, with species mean data suggesting that increases in neuronal density in V1 outpace increases in glial cell density. Primates, however, show a derived scaling relationship, likely due to substantial variation between taxonomic subgroups. Furthermore, apes (including humans) display a higher ratio of glia to neurons than expected based on scaling expectations for Old and New World monkeys; and within the apes, humans display distinctive patterns of cellular densities. While relative increases in glia to neurons in a larger cerebral cortex may often relate to the energetic costs of maintaining larger dendritic arbors and long-range projecting axons that are required, we propose that this relationship does not necessarily hold when considering a limited cortical region or restricted phylogenetic range. As such, we suggest that deviations in cellular densities observed in certain primate lineages represent the evolution of specialized projection layers in V1 along those lineages, and that the relative increase in glia to neurons in the human V1 may represent an evolutionary adaptation for increased metabolic activity among tightly packed populations of granule cells adapted for rapid, achromatic visual awareness. Finally, based on our evidence for an effect of mode of placentation on glia-neuron ratios, we argue that cellular densities relate to the variability of maternal resources to the fetus in a species.

Specializations in primate motion-processing pathways show glia-neuron signatures

Our results suggest that evolutionary variation in the cortical microstructure (e.g., laminar, compartmental,

Table 4 Deviations of predicted from observed values for cell densities in V1 for apes based on OLS regressions generated from plots of Old and New World monkeys ($p < 0.05$)

Species	Glial cell density on neuronal density					
	Observed value	Predicted value	Species mean data		Independent contrasts	
			% Deviation	Within 95% PI	% Deviation	Within 95% PI
Homo sapiens	169,824	83,884	51	No	42	No
Pan troglodytes	144,544	79,432	45	No	29	No
Pan paniscus	138,038	79,238	43	No	30	No
Gorilla gorilla	123,027	63,942	48	No	40	No
Pongo pygmaeus	134,896	66,024	51	No	43	No
Hylobates muelleri	93,325	82,593	12	Yes	5	Yes
Symphalangus syndactylus	101,103	89,839	11	Yes	4	Yes
Macaca fascicularis	100,201	90,002	10	Yes	1	Yes

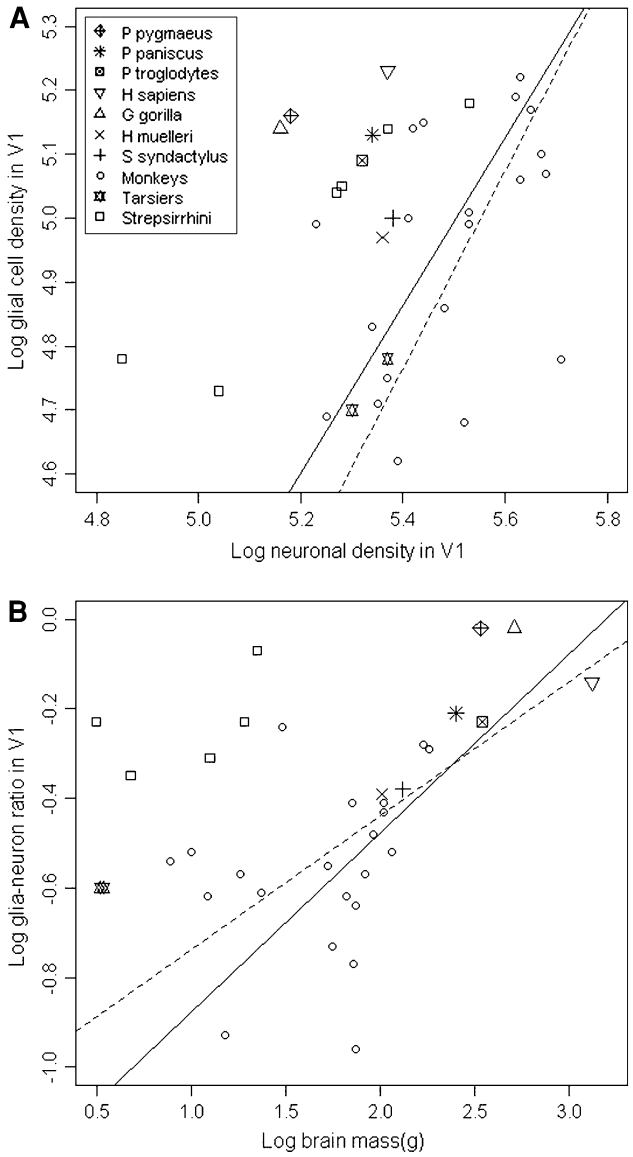


Fig. 4 The allometric scaling of **a** glial cell density on neuronal density and **b** glia-neuron ratio on brain mass for species mean data in primates. **a** The *dotted line* represents LS regressions fitted to Old and New World Monkeys species mean data ($y = 1.54x - 3.55$, $R^2 = 0.531$, $p = 0.011$); the *black line* represents LS regressions fitted to independent contrasts mapped back into tip species space ($y = 1.31x - 2.21$, $p = 0.018$), calculated to predict hypothetical species points attached to the branch leading to apes by pruning apes from the tree and rerooting it at the last common ancestor of apes and other primates. **b** Significant scaling exponents are presented for *Anthropoidea* for species mean data ($y = 0.301x - 1.04$, $R^2 = 0.650$, $p = 0.000$) and independent contrasts ($y = 1.31x - 2.11$, $p = 0.002$)

connectional) of V1 displays phylogenetic diversity that is represented by quantitative changes in the cellular scaling of glia and neurons, which may themselves be considered evolutionary adaptations. Statistically significant differences in the scaling exponents of glia to neurons support three evolutionary events along the primate phylogenetic tree: the divergence of *Strepsirrhini* and *Haplorrhini*, the divergence of apes from Old and New World monkeys, and the divergence of humans from other apes. SMA scaling exponents for independent contrasts in *Haplorrhini*, but not in *Strepsirrhini*, deviate from the scaling exponent generated for eutheria. As such, it is more parsimonious to say that *Strepsirrhini* represents the ancestral primate condition, although it is also likely that both suborders have deviated from the ancestral condition with increased visual specialization.

In *Strepsirrhini*, glial cell density scales roughly isometrically with neuronal density. This is different from the pattern observed in Old and New World monkeys, but consistent with the relationships found in eutheria. Therefore, it appears there may have been little pressure along the strepsirrhine lineage to relax ancestral mammalian constraints on glia-neuron scaling in V1. The reliance of strepsirrhine visual acuity on rods, as well as the reduced necessity of long-range X-related thalamic inputs from the LGN in nocturnal species, may have displaced the need for changes in glia-neuron metabolic coupling for adequate visual processing, although in our analysis the nocturnal owl monkey (*Aotus trivirgatus*) was not shown to deviate from expectations based on *Haplorrhini*.

The phylogenetic affinity for the family *Tarsiidae* has been debated for over a century (Hartwig 2002). To determine whether tarsiers are better affiliated with *Strepsirrhini* or *Anthropoidea* in the context of glial cell and neuronal densities in V1, we plotted the observed values for *Tarsius bancanus* and *Tarsius syrichta* against predictions for both species based on log–log LS regressions of *Strepsirrhini* and Old and New World monkeys. Although observed values for glia-neuron ratio fell closer to prediction values based on brain mass in *Strepsirrhini* than in Old and New World monkeys, glial cell density on neuronal density scaled with allometric expectations based on Old and New World monkeys, rather than *Strepsirrhini*. Therefore, while the tarsier V1 may have been impacted by certain life-history or anatomical adaptations associated with *Strepsirrhini* (e.g., nocturnality, short- and mid-wavelength cone distribution, and small body mass), the molecular mechanism constraining the relationship of glia to neurons observed in Old and New World monkeys seems most likely to have been established at the anthropoid stem lineage.

In *Haplorrhini*, glial cell density scaled significantly with neuronal density in V1 for species mean data and independent contrasts only when apes were not considered, suggesting an evolutionary shift in glia-neuron scaling in V1 at the divergence of apes and other haplorrhines. The novel honeycomb-like geometric arrangement of P- and M-related dendrites in layer IVA of Old and New World monkeys that is conspicuously absent in great apes (Preuss

Table 5 Deviations of predicted from observed values for cell densities in V1 for humans based on LS regressions of non-human apes ($p < 0.05$)

Dependent variable	Independent variable	Observed value	Predicted value	Species mean data		Independent contrasts	
				Deviation (%)	Within 95% CI	Deviation (%)	Within 95% CI
Glia-neuron ratio	Brain mass	0.72	1.16	−60	No	−57	No
	V1 volume		0.74	−3	Yes	2	Yes
	Body mass		0.74	−3	Yes	2	Yes
	Gestation		0.67	8	Yes	10	Yes

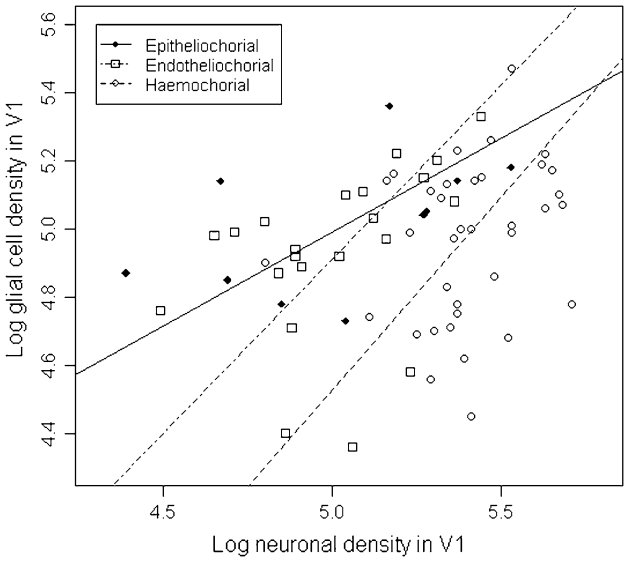


Fig. 5 Allometric scaling of glial cell density on neuronal density reveals a significantly different SMA slope ($p < 0.01$) and y-intercept ($p < 0.01$) for epitheliochorial ($y = 0.553x + 2.24$, $R^2 = 0.468$, $p = 0.028$) compared to endotheliochorial ($y = 1.023x - 0.191$, $R^2 = 0.406$, $p = 0.033$) and haemochorial ($y = 1.133x - 1.12$, $R^2 = 0.565$, $p = 0.019$) species mean data. Slope differences are shown to hold for independent contrasts (Table 6), indicating that the differences in scaling relationships are functional rather than due to phylogenetic relatedness

et al. 1999) may have been concurrent with the observed shift in glia-neuron scaling. The loss of a P-geniculate projection to layer IVA in great apes also may have been accompanied by a concomitant shift in neuronal activity in V1.

While the observed human values for glial cell density on neuronal density fell outside the 95% PI based on a LS regression of Old and New World monkeys, the observed glia-neuron ratio in humans fitted well with predicted values based on brain mass, and much better, in fact, than with the predicted values based on a LS regression of non-human apes. Observed human values for glia-neuron ratio were lower than expected based on scaling exponents for brain mass and gestation length, but within 95% PI based on body mass, for non-human apes. A hypothetical species affixed to the hominoid phylogenetic tree with the observed human glia-neuron ratio we calculated to weigh 53 kg, gestate for 245 days, and have a brain weighing 318 g, showing that the human V1 deviates from the ancestral hominoid pattern of glia-neuron scaling. The presence of a strong M-geniculate projection to layer IVCα in humans (Wong-Riley et al. 1993), which is not found in any other great apes (Preuss and Coleman 2002), may be the reason for the observed increase in neuronal activity. Furthermore, comparative studies of gene expression and cerebral blood flow indicate exceptionally high metabolic activity in the human neocortex. High-density oligonucleotide arrays have identified many genes differentially expressed in the human brain (Caceres et al. 2003; Uddin et al. 2008), which predicts levels of neuronal activity to be exceptionally high in the human neocortex as a result of a considerable upregulation of total gene expression in human evolution (Dicicco-Bloom et al. 1998; Suh et al. 2001; Ferland et al. 2004; Vallender et al. 2008). The possibility that the human neocortex has a uniquely high metabolism is further supported by evidence that, despite trends for larger brains to have lower metabolic rates (per unit volume of tissue) than smaller brains (Aiello and Wheeler 1995), conscious-state cerebral glucose metabolic rates in the human brain (Bohnen et al. 1999; Bentourkia et al. 2000) are as high or higher than those in macaques (Cross et al. 2000; Noda et al. 2002). The upregulation of genes related to lipid metabolism and the importance of dietary docosahexaenoic acid to Na^+/K^+-ATPase activity (Bourre et al. 1989; Djemli-Shipkolye et al. 2003) may be indicative of the importance of dietary changes (e.g., the exploitation of lipid- and docosahexaenoic acid -rich thalassic resources or the scavenging of organ tissue) in early human brain evolution (see Aiello and Wells 2002; Broadhurst et al. 2002; Crawford 2006; Leonard et al. 2007).

Significant molecular changes in the brain may be reflected in corresponding compositional or connectional changes. There is evidence that biochemical mechanisms of generating energy have been under evolutionary selection as early as the anthropoid stem (Grossman et al. 2001, 2004), when brain size began increasing in primates above mammalian predictions. Considering together the consistency of human glia-neuron ratio with predictions based on overall brain mass and V1 volume in Old and New World monkeys, the deviation of human glial cell density from predictions

Table 6 Slope estimates and correlation coefficients in placental groups for cell densities in V1 and anatomical variables

Taxa	Dependent variable	Independent variable	Species mean data					Independent contrasts		
			RMA	R^2	Lower 95% CI	Upper 95% CI	p	RMA	R^2	p
Epitheliochorial ($n = 10$)	Glia cell density	Neuronal density	0.553	0.468	0.307	0.994	0.028	−0.229	0.859	0.000
	Glia-neuron ratio	Brain mass (g)	0.184	0.954	0.117	0.291	0.001	0.172	0.899	0.175
		Body mass (kg)	0.273	0.856	0.165	0.451	0.003	0.195	0.873	0.241
		Gestation length (days)	0.238	0.881	0.141	0.402	0.046	0.149	0.919	0.317
		EQ	−0.183	0.925	−0.106	−0.314	0.714	−0.139	0.934	0.289
Endotheliochorial ($n = 24$)	Glia cell density	Neuronal density	1.023	0.406	0.684	1.532	0.033	1.216	0.346	0.043
	Glia-neuron ratio	Brain mass (g)	0.375	0.632	0.254	0.552	0.020	0.504	0.436	0.012
		Body mass (kg)	0.273	0.788	0.185	0.403	0.023	0.329	0.712	0.022
		Gestation length (days)	0.912	0.013	0.639	1.301	0.002	1.187	0.230	0.128
		EQ	−1.223	0.040	−0.789	−1.898	0.239	−1.280	0.060	0.566
Haemochorial ($n = 32$)	Glia cell density	Neuronal density	1.133	0.565	0.742	1.495	0.019	1.240	0.456	0.003
	Glia-neuron ratio	Brain mass (g)	0.304	0.745	0.221	0.416	0.319	0.345	0.640	0.111
		Body mass (kg)	−0.235	0.803	−0.164	−0.336	0.078	0.306	0.690	0.132
		Gestation length (days)	0.452	0.533	0.335	0.609	0.352	0.641	0.211	0.158
		EQ	−0.248	0.824	−0.181	−0.340	0.031	−0.483	0.402	0.969

based on neuronal density in Old and New World monkeys, and recent evidence that the human V1 has a novel mesh-like arrangement of M-geniculate-related dendrites in layer IVA (Preuss et al. 1999; Preuss and Coleman 2002) and considerable up-regulation of genes in the neocortex (Caceres et al. 2003), it is possible that cortical specializations of the motion-processing pathway in humans occurred without dramatic changes in overall brain or even V1 size.

The diverse cellular scaling relationships among primate taxa, human divergence from glia-neuron ratios based on hominoid predictions, and increased neuronal density based on glial cell density scaling in *Haplorrhini* emphasize phyletic variation in the evolution of V1, identify changes in glial cell and neuronal densities as significant evolutionary adaptations, and support the claim that humans have evolved differentially higher requirements for metabolic support in the neocortex.

Placentation and the cellular brain in eutherian evolution

Associations between the brain and placenta may shift, over evolutionary time, if the two systems are functionally integrated (Walker 2007; Collar et al. 2008; Calsbeek et al. 2009) or genetically correlated (Lande and Arnold 1983; Estes and Arnold 2007; Revell and Harmon 2008). Comparisons of annotated mammalian genes show that, during adaptive radiations, genes involved in reproduction in placental mammals have been under stronger selection than genes involved in cognition (Nielsen et al. 2005; Drake 2007; Rawn and Cross 2008; Gonzalez-Voyer et al. 2009). So if variability in placentation is adaptive in the sense that different modes of placentation affect metabolic investment in the developing fetal brain, then mode of placentation should help explain interspecific variation in cortical organization. One link between placentation and fetal brain development that may be of evolutionary significance is leptin, a protein hormone produced by adipose tissue and placenta. Leptin has been shown to be important for the regulation of maternal energy balance (Zhang et al. 1994; Pelleymounter et al. 1995; Mounzih et al. 1998; Trayhurn et al. 1998), but also for normal neuronal and glial maturation in the developing fetal brain (Ahima et al. 1999). As placental leptin expression and secretion are species-specific traits that may have been adapted for different levels of adiposity during gestation in different species (Zhao

et al. 2003), the diversity of fat storage necessary for different modes of placentation may have influenced selection for increased (or decreased) leptin expression and, therefore, the distribution of neurons and glia in the developing cerebral cortex. More generally, we should expect cellular densities to relate to the variability of maternal resources to the fetus in a species (Martin 1981; Elliot and Crespi 2009). Our data support this prediction. The SMA scaling exponent for glial cell density on neuronal density in the epitheliochorial group, in which the fetus has only indirect access to the maternal blood supply (Mossman 1987), is hypometric for species mean data and negative for independent contrasts, whereas the slopes for independent contrasts in the endotheliochorial and haemochorial species (Fig. 5), in which the fetus has more direct access to the maternal blood supply than species in the epitheliochorial group, scale with positive allometry. However, it is more than simply the invasiveness of the placenta, defined by the number of layers between fetal and maternal blood flow, that determine the efficiency of exchange or relative availability of resources in the placenta (Enders and Carter 2004), as recently described patterns of association between maternofetal interdigitation and interhemal barrier along eutherian lineages have indicated that evolution of placental invasiveness was not principally driven by a need for more efficient oxygen exchange (see Wildman et al. 2006; Lewitus and Soligo 2011). The significant predictive power of gestation length for glia-neuron ratio in eutheria (t value $= 4.729$, $p = 0.000$) and carnivores (t value $= 2.976$, $p = 0.008$), and the significantly different scaling coefficients for cellular densities among the placental groups indicate that placentation affects the relationship of cortical glia to neurons in the evolving mammalian brain.

Conclusions

The phylogenetic diversity of glia-neuron ratio in our data set highlights the difficulty of using brain mass as an indicator of cellular densities. We propose that such diversity is predicated on both cellular mechanisms acting on energy metabolism and behavioral adaptations affecting visual cortical processing, and that each adaptation promotes a quantifiable evolutionary signature in neuronal and glial cell density.

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